BOOK III — QUANTUM REALITY IN GRUT

> **WORKING DRAFT — part of the GRUT working corpus; statuses per `books/CORPUS_CHARTER.md`; subject to chapter-by-chapter audit; nothing here banks.**

GRUT Books — Working Edition v1.0 · D. Ryan Grover · 2026-09-07 · release `zenodo-v1.0` · Markdown is the authoritative source; the PDF is generated from it. Drafted from the frozen record by the RAI builder (Claude, Anthropic) under owner direction — this edition is a publishing pass only: formatting, navigation, and metadata; no claim strengthened or weakened for publication.

Contents

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1 · Scope, and the reading rule this book lives or dies by

This book presents what the GRUT record says about quantum mechanics: how the framework recovers standard quantum theory, what its open-system machinery does for state evolution and decoherence, where the Born probability structure comes from, what its one proposed quantum-gravitational observable is, and what its cleanest theorem-grade result forbids.

Four different kinds of statement live in these pages, and the entire value of the book is that the reader can never mistake one for another:

- **Recoveries** — standard quantum physics reproduced on GRUT's declared inputs, largely *by identity*: the executed machinery is standard machinery. Recovery establishes compatibility, not correspondence evidence, and it earns nothing toward prediction.
- **Assumptions** — declared inputs, priced on the register's ledger. The most important one in this book, the Born measure, is the framework's openly flagged never-sorted door.

- **Hypotheses** — proposed relations, not derived, not validated. Exactly one lives here: the rung8 gravitational-decoherence shape signature.
- **One derivation** — the linear-universe response no-go from the RRT arm, which is a theorem about *all* linear dynamics and therefore, precisely because of its generality, carries no confirmatory weight for GRUT. It closes a door rather than opening one.

The owner's standing directive governs throughout: no invented derivations, no promotion of hypotheses to predictions, no reopened gates. Where the record is silent — and on several questions a physicist would ask first, it is silent — this book says so explicitly. The PREDICTED set is empty across the corpus, and nothing in this book is an exception.

> **STATUS: EMPTY (nothing has earned entry; Book IX governs entry)** — canonical claim 21; the PREDICTED set contains no quantum-sector entry (source: `GRUT_PROGRAM_FREEZE.md` §3, `books/CORPUS_CHARTER.md`).

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2 · The backbone: one action, one reduction

GRUT's quantum story starts where the whole framework starts: a single Schwinger–Keldysh influence action for the system degrees of freedom against a declared bath,

```
S_IF[h_r, h_a] = ∫ (dω/2π) d³k [ h_a · K_R(ω,k²) · h_r + (i/2) h_a · N(ω,k²) · h_a ]
```

with K_R the retarded dissipation kernel and N the noise kernel (`S_IF.md`, `provenance/claims.json#rung1_inin_formalism`). Everything quantum-mechanical in this book descends from one move: **integrating out the declared bath yields the reduced dynamics of the system's density matrix** — the reduced master equation, in the Feynman–Vernon / Calzetta–Hu sense the register cites (`feynman_vernon1963`, `calzetta_hu_book` in `provenance/sources.json`).

Two honesty notes attach to the backbone before any physics is read off it.

First, the form itself proves nothing. The influence-functional form is universal: any local, causal open quantum system coarse-grains to S_IF = K_R + (i/2)N. GRUT adopts a universal IR language; it does not own the universality.

> **STATUS: RECOVERED (generic; u1: the form confers no GRUT-specific content)** — canonical claim 1 (source: `provenance/claims.json#u1_form_universality`, `GRUT_MODEL_FRAMEWORK.md` §3).

Second, writing S_IF at all costs declared inputs. The register prices four on the formalism node alone: the system/bath split, the Gaussian/linear-response truncation, the background Lorentzian causal structure, and the 4d-covariant availability of the Ward-sourced gauge-orbit zero (booked 2026-08-17, discharge condition named).

> **STATUS: ASSUMPTION (four declared inputs, +4 on the rung1 formalism node; the system/bath split is an AXIOM-form input whose specific partition is a structural selection)** — source: `provenance/claims.json#rung1_inin_formalism` ledger_note; `GRUT_MODEL_FRAMEWORK.md` §2.

In equilibrium the noise kernel is not free: N is locked to Im χ by the fluctuation–dissipation theorem with the coth($\hbar\omega$/2kT) factor, and KMS detailed balance is enforced on every admissible kernel as a hard admission gate (`gate/kms.py`). This is the one place the ledger *removes* an input (−1): N ceases to be independent.

> **STATUS: ASSUMPTION (borrowed standard identity, enforced as a hard admission gate; rung2) —** canonical claim 2 (source: `provenance/claims.json#rung2_kms_gate`; `callen_welton1951`, `kubo1966` per `provenance/sources.json`).

Standard-physics background, distinguished from GRUT's record: the reduction from a closed system-plus-bath to a closed dissipative master equation is the Nakajima–Zwanzig / Born–Markov construction, and it consumes inputs of its own — most notably the factorized initial system–bath state (the quantum Stosszahlansatz: the closed dissipative master equation is literally the $Q\rho(0)=0$ deletion of initial correlations) and the continuum limit of bath modes. The GRUT record does not hide this; its own arrow-of-time analysis (`calc/RESULTS_arrow.md`, `docs/WHERE_IT_STOPS.src.md` §I.2) identifies the factorization as one of the three interchangeable guises of the imported past-boundary condition. The direction of relaxation in the reduced dynamics is Book V's subject; what matters here is that the master equation's irreversibility is *supplied by declared inputs*, not generated by the formalism, and the record says so on its face.

What the record does **not** contain is a GRUT-specific master equation for a concrete laboratory system, derived in-house with novel structure. The reduction is cited machinery on declared inputs. The one executed in-house master-equation computation in the quantum sector is the rung8 energy-basis decoherence-rate calculation of §7 — and it is a magnitude audit of a hypothesis, not a new dynamical law.

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3 · The unitary core: Schrödinger recovered

Switch the noise off and the reduced dynamics closes on the system: the influence-functional reduction reproduces the Schrödinger equation as the unitary core of the QM limit. This is rung6's first clause — "integrating out the bath yields the reduced-density-matrix master equation; unitary core = Schrödinger" — and it is a recovery in the strict, deflationary sense this corpus uses everywhere: the machinery that produces it is standard open-system machinery, and the result establishes that GRUT *contains* quantum mechanics, not that it *explains* it.

> **STATUS: RECOVERED (noise-free limit of the influence-functional reduction; the register carries the composite rung6 node at tier `assumed` with +2 declared imports — a recovery-with-imports, same shape as the GR limit) —** source: `provenance/claims.json#rung6_qm_limit`; frozen summary `GRUT_MODEL_FRAMEWORK.md` §5.

The two imports are the content of the "assumed" tier, and the register declares them by name (rung6 ledger_note, +2):

1. **The quantization / single-valuedness condition.** The record books this as declared import (i) and nowhere elaborates or derives it; it is a line-item on the ledger, not a worked document. This

book records the absence as an absence.

> **STATUS: ASSUMPTION (declared import (i) on rung6; carried on the ledger, not elaborated anywhere in the record)** — source: `provenance/claims.json#rung6_qm_limit` ledger_note.

2. **The Born probability-measure postulate** — §5 of this book, where it gets the full treatment it is owed.

A discipline note the record keeps on its own face: the rung6 ledger was once undercounted as +1 — one of the two imports was being carried silently — and the kill-shot audit that caught it is recorded in the node itself. The corrected +2 stands, with a written stance justification under the register's `laundering_ok` waiver: the waiver says the ledger, not the tier, carries the inputs.

The node's own overturning computation states what would break the recovery: show that the influence-functional reduction does *not* reproduce Schrödinger in the noise-free limit — or show that decoherence alone yields the Born measure (it does not; the improper-mixture objection, §5).

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4 · Decoherence: the noise kernel and the pointer basis

With the noise on, the second half of rung6 engages: **the noise kernel N supplies decoherence, and decoherence selects a pointer basis.** Superpositions of states that the bath distinguishes lose their mutual coherence at a rate set by N; the surviving basis is the one the system–bath coupling picks out. In GRUT's telling this is a sector instance of the one architecture — the same N that the KMS gate locks to Im χ in equilibrium is the object doing the einselecting.

> **STATUS: RECOVERED (standard decoherence machinery on declared inputs)** — canonical claim 11 (source: `provenance/claims.json#rung6_qm_limit`; `NO_GO_LEDGER.md` entry 7; `GRUT_MODEL_FRAMEWORK.md` §5).

Two grading remarks, both from the register:

- **The decoherence *rate* is the genuinely differentiating output.** The rung6 node's differentiator field is explicit: "Decoherence RATE is differentiating; Born rule FAILS-DIFFERENTIATION (inherited postulate)." A bath with GRUT's declared kernels produces specific rates, and rates are in principle measurable — which is exactly why the program's one proposed quantum observable (§7) is a decoherence-rate structure and not anything deeper.
- **Basis selection is not outcome selection.** This is the load-bearing boundary of the whole chapter, and the record draws it in one sentence (`NO_GO_LEDGER.md` entry 7): the reduction reproduces the Schrödinger core plus environment-induced decoherence that selects a pointer *basis* — but a preferred basis is **not** outcome selection. That boundary is §5.

The record contains no in-house pointer-basis computation for a concrete system — no worked model exhibiting einselection dynamics from GRUT's specific kernels, beyond the energy-basis rate analysis of §7. The pointer-basis claim is cited standard machinery (Feynman–Vernon; Calzetta–Hu) applied to declared inputs, and this book grades it exactly so: RECOVERED, not DERIVED.

5 · The Born measure: the never-sorted door

Here is the chapter the corpus is contractually obliged to write honestly, because the temptation to blur it is the oldest failure mode in the decoherence literature.

Decoherence, run to completion, delivers a reduced density matrix diagonal in the pointer basis. What it does **not** deliver is the probability that any particular diagonal entry is *the outcome*. GRUT does not produce the Born measure. It inherits it.

> **STATUS: ASSUMPTION (inherited axiom; the improper-mixture objection stands)** — canonical claim 10 (source: `provenance/claims.json#born_rule`, `provenance/claims.json#rung6_qm_limit`, `NO_GO_LEDGER.md` entry 7, `POSTULATE_MAP.md` Bin 1 item 3).

The improper-mixture objection, as the record states it and as standard-physics background fills it in: a reduced density matrix that is diagonal in some basis is an *improper* mixture — it arises from tracing out entanglement, not from classical ignorance over outcomes that have already happened. Reading its diagonal entries as probabilities of single outcomes presupposes exactly the probability interpretation one was trying to derive. Decoherence plus a pointer basis is *necessary but not sufficient*; the $|\psi|^2$ weighting must come from elsewhere (`NO_GO_LEDGER.md` entry 7, "spec for a completion").

The register's bookkeeping around this is unusually careful and worth reproducing, because it is the difference between hiding an assumption and displaying one:

- The Born rule is a **first-class register node** (`born_rule`, tier `assumed`, grut_standing `borrowed`), promoted 2026-07-06 precisely so the borrow would be *visible and edged*: rung6_qm_limit formally `depends_on` born_rule. An import you can see in the dependency graph cannot be laundered.
- The node carries **ledger_delta 0** — not because it is free, but because its +1 cost is already counted inside rung6's declared +2; carrying it twice would double-count, and the register's blind-sum auditor cannot catch double counts, so the humans did.
- Its overturning computation is stated: a derivation of the Born rule *from* S_IF without assuming it would promote borrowed → derived. "None exists; GRUT imports it."
- `POSTULATE_MAP.md` sorts it into **Bin 1 — irreducible primitives**: not an open layer with a path to derivation, but bedrock, alongside the responsive-medium ontology and the Past Hypothesis. "Anything claiming to derive one of them is laundering."
- The program freeze flags it with a distinct shade of honesty: the Born measure "never once entered any campaign's sort" (`GRUT_PROGRAM_FREEZE.md` §3) — every other primitive was attacked, audited, or triangulated at least once; this one was declared and then left standing. It is the framework's never-sorted door, and the freeze says so.

One meta-note completes the picture. The freeze's own falsification section (`GRUT_PROGRAM_FREEZE.md` §6) lists "a demonstrated second discharge row anywhere (einselection, Born-measure sort, CLPW outcome A)" as something that would *break* the consolidated record's conservation pattern — the finding that every closure consumed an input. A successful derivation of the Born measure from decoherence would therefore not merely upgrade one node; it would overturn the program's central consolidated claim. The record has priced its own refutation.

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6 · Measurement and interpretation: the absence map

What does GRUT say about the measurement problem? The honest answer, and the record's own answer, is: **it inherits it, unsolved.** "GRUT recovers quantum mechanics as a limit, not a derivation: the rate is earned, the Born rule is borrowed. The measurement problem is inherited, not solved" (`NO_GO_LEDGER.md` entry 7, calibration).

The absence map, itemized — each line an explicit statement that the record is silent, which the charter counts as valid content:

- **No interpretation is adjudicated.** The record contains no node, computation, or ruling selecting Everett, collapse, pilot-wave, relational, or any other interpretation of the quantum state. The words do not appear as claims anywhere in the register.

> **STATUS: UNMAPPED (the record contains no GRUT account of interpretation selection)** — source: absence verified against `provenance/claims.json` (74 nodes) and the repository's consolidation documents.

- **No collapse mechanism is proposed.** GRUT's noise kernel decoheres; it does not collapse. The framework's one contact with the collapse-model literature is *adversarial*: the rung8 signature is defined by its *distinction* from Diósi–Penrose and CSL (§7), not by membership in that family.
- **No Bell-inequality, contextuality, or entanglement-structure account exists.** No node covers nonlocal correlations, CHSH structure, or entanglement thermodynamics.

> **STATUS: UNMAPPED** — source: absence verified by repository-wide search; nearest contact is the Gisin-signalling caution attached to nonlinear extensions in `rrt0/RRT1_REDUCIBILITY_CORRECTION.md` §5 (branch `rrt0-phase2`), which is a fence, not an account.

- **The double-slit founding hypothesis is deferred as analogy-only.** The founding-era hope that GRUT yields a unique double-slit observable (H3) was screened 2026-06-27: the original two-state mapping had no operator, coupling, or observable behind it — standard complementarity accounts for everything invoked — and the one steelman (Casimir-modified interferometry) is killed by GRUT's *own* rung8 result: the vacuum coupling decoheres in the energy basis (the wrong basis for a double-slit), samples $S(0) = 0$, and the surviving wedge is 7–47 orders below detectability. H3 becomes a GRUT claim only if a unique observable distinct from standard QM is exhibited; none has been.

> **STATUS: UNRESOLVED (deferred; analogy-only absent a unique observable; register tier `to-derive`, disposition deferred)** — source: `provenance/claims.json#founding_h3_doubleslit_anchor`.

A vocabulary annotation attached to that node binds this whole corpus: the "specialist," "referee," and "overseer" passes recorded across the register are in-house AI sessions run by the program's own human author from clean contexts; no outside human has ever been contacted by this program. The register carries the annotation rather than renaming the history — the drift is itself the finding.

What GRUT *positively* offers the measurement discussion is exactly one thing: a unified home for the decoherence half. The same N, the same KMS lock, and the same passivity structure that govern dissipation and gravitational response also govern which basis survives monitoring — one architecture across sectors, with the outcome problem left standing exactly where standard decoherence theory leaves it.

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7 · Gravitational decoherence: the rung8 hypothesis

This is the book's one hypothesis, and it must be read with its label welded on. The record calls it the USL shape signature (the register's name for the rung8 gravitational- decoherence signature; the record does not expand the acronym).

The proposal. Take GRUT's noise kernel N and let it drive the Anastopoulos–Hu (2013) gravitational-decoherence master equation (arXiv:1305.5231, the record's cited source). The result is a decoherence signature distinguished from Diósi–Penrose and CSL **in shape only**: AH-type decoherence acts in the **energy basis**, whereas DP/CSL localize in **position**. Born–Markov reduction of the influence action gives, for coherence between energy eigenstates split by ΔE (`calc/RESULTS_energy_basis.md`):

```
Γ(ΔE) = (1/ħ²) |A_nm|² S(ΔE/ħ)
```

— the rate samples the vacuum noise spectrum at the Bohr frequency of the energy gap, with a predicted parameter-free *shape* g(ΔE): suppressed rise, peak at $\Delta E = 1.22\ \hbar\omega_c$, FWHM $\Delta E \in [0.69, 1.85]\ \hbar\omega_c$, then cutoff. The experimental wedge is orthogonal by construction: vary ΔE at fixed Δx and GRUT-AH responds while DP/CSL stay flat; vary Δx at fixed ΔE and DP/CSL respond while GRUT-AH stays flat.

> **STATUS: HYPOTHESIS (proposed phenomenological relation; shape-only; magnitude verdict quiet-or-faint)** — canonical claim 12 (source: `provenance/claims.json#rung8_falsifier`, tier `to-derive`; `calc/RESULTS_energy_basis.md`; `GRUT_MODEL_FRAMEWORK.md` §5).

What is staked, not predicted. The register declares +2 inputs on the node: (i) the overall amplitude/coupling normalization κ, which must survive the MICROSCOPE/LISA-Pathfinder/Donadi bounds, and (ii) the cutoff scale ω_c that places the energy-gap peak — which is what the retired "689 Hz, parameter-free" claim really was. The shape and the energy-vs-position wedge are proposed as consequences; the amplitude and the peak location are staked.

> **STATUS: ASSUMPTION (two staked inputs, +2: amplitude κ and cutoff ω_c; the '689 Hz parameter-free' billing is RETIRED on the record)** — source: `provenance/claims.json#rung8_falsifier` ledger_note; `calc/RESULTS_energy_basis.md`.

The magnitude verdict — quiet-or-faint. The Q1 computation (`calc/q1_energy_basis_magnitude.py`, 2026-06-25) asked the decisive operator-algebra question: does the effective gravitational coupling A carry nonzero Bohr-frequency components, [A, H_S] ≠ 0? The answer, computed ratio-first: the *dominant* coupling is the diagonal T^{00} ∼ energy density, which commutes with H_S and therefore samples S(0) = 0 — the responsive vacuum is a **quiet** bath for static energy superpositions (Γ = 0). The wedge-carrying off-diagonal couplings T^{0i}/T^{ij} (∼v/c) survive but land **7–47 orders of magnitude below current sensitivity**; observability would require staking the noise amplitude $\sim 10^7$ or more above its natural value — a tuned number at the current matter-wave bound. The robust Pikovski time-dilation decoherence does work, but it is position-basis — the same axis as DP/CSL, not the wedge. The falsifier does not carry the program.

> **STATUS: CLOSED (computed magnitude verdict quiet-or-faint — the no-go ledger's INVISIBLE-BY-SUPPRESSION strength; the node itself stays register-tier `to-derive` with a named gray-zone check pending: any leading-order off-diagonal energy coupling sampling S(ΔE) at O(1)?)** — source: `provenance/claims.json#rung8_falsifier` tier_note and differentiator; `NO_GO_LEDGER.md` entry 4; instruments `calc/q1_energy_basis_magnitude.py`, `calc/energy_basis_decoherence.py`.

Scope fences the record itself carries, reproduced here because they travel with the claim:

- *Cutoff convention (Ruling A, 2026-08-23):* the calculation uses the staked tabletop cutoff ω_c = 2π·689 rad/s; there is no universal ω_c — every calculation must name its cutoff; spectral claims are scoped to ω ≪ ω_c, and the vicinity ω ∼ ω_c is outside the declared validity domain.
- *The spectral shape rides on an open anchor.* The finite-T shape check (`calc/finite_T_exponent.py`: S(ω) analytic at ω = 0, no second slow pole within the analytic class) is confirmed, conditional on the analytic class holding and no second bath scale — but the pole-vs-cut anchor itself is open. The qualitative energy-vs-position wedge never depended on it; the quantitative shape does.

> **STATUS: UNRESOLVED (anchor-class, derived-pending; pole-vs-cut open; the Tier-4 computation found a CUT, not a pole, at flat scope)** — canonical claim 4 (source: `provenance/claims.json#rung3_single_pole`).

- *The BMV entanglement-witness backup is WITHDRAWN* — an energy-basis decoherer may not degrade a position-basis witness; recompute or drop. It has not been recomputed.

The standard the shape-claim must eventually meet. The record's own structural-theory adjudication (`RAI_STRUCTURAL_THEORY_SEARCH.md` §6) identifies the one historical instance of a foundations-level structural claim killed by an experiment: the **parameter-free Diósi–Penrose model, excluded at Gran Sasso** — the template, not the outlier (`GRUT_MODEL_FRAMEWORK.md` §5 carries the same context line). That is the bar. DP in its parameter-free form made a magnitude claim and died by it; GRUT's rung8, in its current quiet-or-faint state, makes a shape claim whose magnitude sits 7–47 orders below reach and whose amplitude is staked, not derived. Until the gray-zone check or a new mechanism lifts the magnitude into a distinguishable regime, the USL signature is a hypothesis that cannot yet be killed the way DP was killed — which, by the program's own lights, is a *deficit*, not a safety. The framework's weakening conditions (`GRUT_MODEL_FRAMEWORK.md` §8) explicitly include "the USL shape-signature excluded in the regime where it is distinguishable" — the record wants this claim exposed to fire, and it currently is not.

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8 · The linear-universe response no-go (the RRT arm)

The strongest quantum-sector result in the record is a negative one, and it was found by the program's own adversarial arm running against itself. It lives on the sibling branch `rrt0-phase2` (working tree `/Users/mpg/Desktop/testing`), and its provenance discipline is part of its content.

What RRT-0 was. A pre-registered, frozen-before-simulation hostile test (`rrt0/RRT0_SPEC.md`): a d = 4 closed unitary toy universe (GUE Hamiltonian, density matrices, integer update steps), a pre-registered internal intervention family E_α(ρ) = U^τ((1−λ)ρ + λσ_α)U^(−τ), an influence statistic Φ, and frozen decision gates — run under a claim firewall (`rrt0/RRT0_CLAIM_BOUNDARIES.md`) that marks every output UNBANKED / PRE-REGISTERED / EXPLORATORY, **NOT GRUT evidence**, with a CI test that fails any output document claiming more.

What it found. The intervention influence statistic is *fully reducible*: for the registered class, Δρ(t,τ) = U^τ(E[ρ_0(t)] − ρ_0(t))U^(−τ) exactly, residual identically zero (`rrt0/MODEL_CLASS_VERDICT.md` — derived in preflight, registered before any battery run). Raw Φ is a propagation/response diagnostic, not a diagnostic of irreducible emergence.

What the correction made of it. The RRT-1 design audit initially claimed the no-go rested on invertibility and that a general CPTP channel escapes it. Both claims were caught false by the owner and corrected on the record (`rrt0/RRT1_REDUCIBILITY_CORRECTION.md`). The generalized identity is one line from linearity alone: for ANY linear map $\mathscr{T}$ on operators, any intervention E, any step count n,

```
Δ_n := 𝒯ⁿ(E[ρ]) − 𝒯ⁿ(ρ) = 𝒯ⁿ((E−I)[ρ]).
```

The exact load-bearing assumptions: $\mathscr{T}$ linear; E linear or affine; readout linear (Tr[B·]). **Not required:** invertibility, unitarity, trace preservation, complete positivity, a semigroup property, a fixed point. Numerical confirmation at machine zero (residuals ≤ 9.5×10^{-17}) on genuinely non-invertible channels — depolarizing, non-unital amplitude damping, full rank-collapsing reset — and on Lindblad evolution e^{t$\mathscr{L}$} for arbitrary generators.

> **STATUS: DERIVED (RRT arm: intervention response reducible for all linear dynamics; escaped only by nonlinearity)** — canonical claim 20 (source: `rrt0/RRT1_REDUCIBILITY_CORRECTION.md` and `rrt0/MODEL_CLASS_VERDICT.md`, branch `rrt0-phase2`; consolidated in `GRUT_PROGRAM_FREEZE.md` §3 DERIVED list).

The P1–P4 separation — the correction's lasting conceptual contribution, because the design audit had conflated the first two:

proposition	statement	holds iff
P1 response reducibility	$\Delta = \mathscr{T}^n((E-I)\rho)$	$\mathscr{T}$, E, readout linear — invertibility plays NO role

proposition	statement	holds iff
P2 state recoverability	ρ_0 recoverable from ρ_t	$\mathscr{T}$ invertible — this is where invertibility lives
P3 asymptotic-structure reducibility	attractors/pointer/DFS reduce to Fix($\mathscr{T}$) and the peripheral algebra	a function of $\mathscr{T}$'s spectral data
P4 operator-algebra reducibility	emergent preferred algebra = a function of the generator's algebra	governed by Fix/Comm{L_a}

RRT-0 tests P1, and P1 is escaped **only by abandoning linearity** — of the dynamics, the intervention, or the readout. Nonlinear quantum dynamics is non-standard and carries its own no-gos (Gisin signalling), so the escape hatch is named, fenced, and expensive (Fork B); the recommended successor question (Fork A) keeps linearity and asks the P3/P4 question instead, itself predicted to yield another no-go class.

> **STATUS: DERIVED (analytic correction audit: the four-proposition separation and the corrected minimum escape condition — loss of linearity, not invertibility)** — source: `rrt0/RRT1_REDUCIBILITY_CORRECTION.md` §§1–4.

What this means for quantum reality in GRUT. Three things, in descending order of comfort:

1. **Every "emergence" reading of intervention response within linear quantum mechanics is closed.** Any claim that a linear open quantum system — unitary, dissipative, CPTP, Lindblad, GRUT's reduced dynamics included — exhibits intervention-response structure beyond what its supplied $\mathscr{T}$, E, and readout encode, is false by identity. The program's own P1 emergence framing is listed among the killed in `GRUT_PROGRAM_FREEZE.md` §4.
2. **This is deflationary for GRUT, and the record insists on that reading.** The no-go is broader than designed — a property of the entire linear dynamical universe — so it confers no distinction on GRUT. What it forbids, it forbids for everyone; what it leaves, it leaves for everyone. The claim-boundary firewall (NOT GRUT evidence) and the freeze's consolidation coexist without tension: the theorem is banked as *mathematics the program produced*, not as *support for the program's physics*.
3. **What cannot be concluded** is fenced in the correction itself: not that open systems lack interesting structure (P3 attractors, pointer states, and decoherence-free subspaces all exist — they are functions of the supplied $\mathscr{T}$); not that nonlinearity *would* yield irreducible organization (unknown; only that it is the sole P1 escape); nothing about the half-line/KMS residue; no transfer of RRT-0's passes to any new class.

The RRT-0 arc's procedural record is also part of this book's evidence about *how* the program treats quantum claims: five consecutive self-refusals on the record — semantic PASS, reducibility PASS, mirror PASS (bit-exact), sector UNRESOLVED, CPR reference ABSENT (6/6 seeds) — followed by the broadening of the no-go (`GRUT_PROGRAM_FREEZE.md` §7; reports under `rrt0/reports/`). Even the final reducibility-gate rerun carries its own correction on its face: a 2026-09-06 audit found the prior gate's independent propagator route mathematically invalid (a Hermitian-only eigensolver applied to a non-Hermitian unitary), superseded it with two valid routes, preserved the failed artifact with zero evidentiary status, and re-passed at max residual 2.7×10^{-14} over 15,120 cases (`rrt0/reports/REDUCIBILITY_GATE.json`, SUPERSEDED_AUDIT block). The instrument corrects itself in public. That habit, not any quantum mechanism, is what the record offers as its contribution to quantum foundations.

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9 · What would move this book

Every claim above names its own overturning condition; collected, they are the book's forward edge — none is scheduled, and none may be presumed:

- **A derivation of the Born measure from S_IF without assuming it** promotes the borrowed node to derived — and simultaneously breaks the freeze's consolidated "every-closure-consumed-an-input" record (§5, §6 above). Highest stakes, no known route.
- **Failure of the Schrödinger recovery in the noise-free limit** would break rung6's recovery clause (the node's own overturning computation). No indication exists.
- **The rung8 gray-zone check**: a leading-order off-diagonal energy coupling sampling S(ΔE) at O(1) would lift quiet-or-faint and make the USL shape signature testable — at which point it faces the Gran Sasso standard. Conversely, exclusion of the shape in a regime where it is distinguishable is a named weakener of the whole framework (`GRUT_MODEL_FRAMEWORK.md` §8).
- **A Fork-B construction** — declared, firewalled nonlinearity in dynamics, intervention, or readout — is the only route by which intervention response can become irreducible. The stopping rule governs: no new generation merely because the previous one failed; an independently motivated principle naming a phenomenon the linear framework cannot represent must open the door (`GRUT_PROGRAM_FREEZE.md` §§1, 5).
- **A unique double-slit observable computable from GRUT** would revive H3 from deferral. None has been exhibited.

Sources drawn from

- `books/CORPUS_CHARTER.md` (status vocabulary; canonical claims 1, 2, 4, 10, 11, 12, 20, 21)
- `GRUT_MODEL_FRAMEWORK.md` (§§2–5, 7–8 — the frozen QM/USL summary this book stays consistent with)
- `GRUT_PROGRAM_FREEZE.md` (§§1–7 — ledger, stopping rule, RRT-0 arc consolidation)
- `provenance/claims.json` — nodes `rung6_qm_limit`, `rung8_falsifier`, `born_rule`, `rung1_inin_formalism`, `rung2_kms_gate`, `rung3_single_pole`, `rung5_gr_limit`, `u1_form_universality`, `founding_h3_doubleslit_anchor`, `emergence_chain`
- `provenance/sources.json` (external citations: `feynman_vernon1963`, `calzetta_hu_book`, `calzetta_hu_ctp1988`, `schwinger1961`, `keldysh1964`, `callen_welton1951`, `kubo1966`, `born1926`, `anastopoulos_hu2013` [arXiv:1305.5231], `bassi_rmp2013`)
- `NO_GO_LEDGER.md` (entries 4 and 7)

- `POSTULATE_MAP.md` (Bin 1 item 3; modular decomposition M4/M6)
- `S_IF.md` (the action, kernel structure, structural conditions)
- `calc/RESULTS_energy_basis.md`; `calc/RESULTS_arrow.md` (master-equation input context)
- `docs/WHERE_IT_STOPS.src.md` (§I.2, Nakajima–Zwanzig factorization input)
- `RAI_STRUCTURAL_THEORY_SEARCH.md` (§6 — the Gran Sasso DP-exclusion template)
- `GRUT_NEXT_STEPS.md` (USL shape-signature forward items)
- Branch `rrt0-phase2` (working tree `/Users/mpg/Desktop/testing`): `rrt0/RRT1_REDUCIBILITY_CORRECTION.md`, `rrt0/RRT0_SPEC.md`, `rrt0/RRT0_CLAIM_BOUNDARIES.md`, `rrt0/MODEL_CLASS_VERDICT.md`, `rrt0/reports/REDUCIBILITY_GATE.json`, `rrt0/reports/MIRROR_CONTROL.json`, `rrt0/reports/SECTOR_SELECTION_FIREWALL.json`, `rrt0/reports/CPR_FEASIBILITY.json`

Gaps in this book

1. **No interpretation of quantum mechanics is adjudicated anywhere in the record —** Everett, collapse, pilot-wave, relational: no node, no ruling. This book reports the silence; it does not fill it.
2. **The quantization/single-valuedness condition (rung6 import (i)) is a ledger line-item only.** No document in the record elaborates what is imported or why it takes that form. An audit of that import is owed and unscheduled.
3. **No in-house pointer-basis computation for a concrete system exists** — einselection is cited machinery; no worked model exhibits basis selection from GRUT's specific kernels.
4. **Bell-inequality structure, contextuality, and entanglement thermodynamics are UNMAPPED** — no GRUT account exists at all.
5. **The rung8 gray-zone check is pending** (off-diagonal energy coupling at O(1)); until it is run, quiet-or-faint is the standing verdict and the USL signature is untestable.
6. **The BMV witness under an energy-basis decoherer was withdrawn and never recomputed.**
7. **The rung8 spectral shape is conditional on the open rung3 anchor** (pole-vs-cut, analytic class, no-second-scale proviso); a cut-class resolution would force the shape claim to be redone.
8. **Fork B (nonlinearity) is named but wholly unexplored —** the only P1 escape has no construction, no model, and a standing Gisin-signalling caution.
9. **The relation between GRUT's reduced dynamics and specific laboratory master equations** (beyond the AH-driven rung8 lead) is not worked anywhere: no GRUT-specific Lindblad coefficients for any real system are on the record.
10. **This book leans on the frozen summaries for cross-sector claims** (arrow/Past Hypothesis → Book V; algebraic/CLPW structure → Book VIII; test governance → Book IX); where those books sharpen statuses, this book's cross-references inherit the audit.